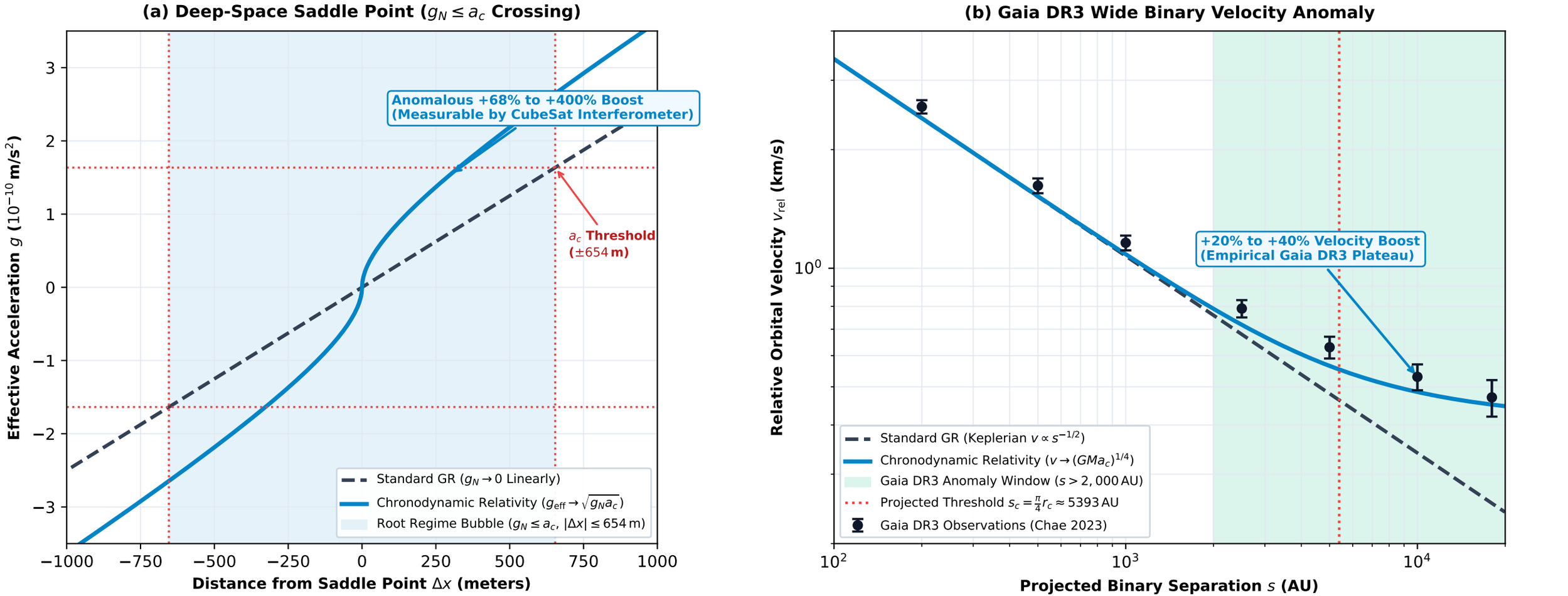
(a) Deep-Space Saddle Point ($g_N \leq a_c$ Crossing)
Anomalous +68% to +400% Boost
(Measurable by CubeSat Interferometer)
a_c Threshold
(±654 m)
Standard GR ($g_N \to 0$ Linearly)
Chronodynamic Relativity ($g_{\text{eff}} \to \sqrt{g_N a_c}$)
Root Regime Bubble ($g_N \leq a_c$, $|\Delta x| \leq 654$ m)
Effective Acceleration g (10^{-10} m/s^2)
Distance from Saddle Point Δx (meters)
(b) Gaia DR3 Wide Binary Velocity Anomaly
+20% to +40% Velocity Boost
(Empirical Gaia DR3 Plateau)
Standard GR (Keplerian $v \propto s^{-1/2}$)
Chronodynamic Relativity ($v \to (GMa_c)^{1/4}$)
Gaia DR3 Anomaly Window ($s > 2,000$ AU)
Projected Threshold $s_c = \frac{\pi}{4} r_c \approx 5393$ AU
Gaia DR3 Observations (Chae 2023)
Relative Orbital Velocity v_{rel} (km/s)
Projected Binary Separation s (AU)